

Excerpts from Chia-Chiunn Ho's recollections of Larry Zitnick's AI class at Facebook

- Researchers like Zitnick were building machine vision one tiny piece at a time, applying very particular techniques to very particular parts of the problem. But then academics like Geoff Hinton showed that a single piece—a deep neural network—could achieve far more. Rather than code a system by hand, Hinton and company built neural networks that could learn tasks largely on their own by analysing vast amounts of data. “We saw this huge step change with deep learning,” Zitnick says. “Things started to work.”
- A neural net mimics the web of neurons in the brain. It operates by sending information between processing units, or nodes, that stand in for neurons. But these nodes are really just linear algebra and calculus that can identify patterns in data.
- Each layer makes a calculation and passes it to the next. Then, using a technique called ‘back propagation,’ the layers send information back down the chain as a means of error correction. As the years went by and technology advanced, neural networks could train on much larger amounts of data using much larger amounts of computing power.
- “For the first time ever, we could take raw input data like audio and images and make sense of them,” Zitnick told his class.
- Convolutional neural networks is a method inspired by the brain’s visual cortex that groups neurons into ‘receptive fields’ arranged almost like overlapping tiles.
- Long short-term memory networks (LSTMs) are neural networks that include their own short-term memory—a way of retaining one piece of information while examining what comes next. This is what helps identify the commands you speak into Android phones.
- In the end, all these methods are still just math. But to understand how they work, students must visualise how they operate across time (as data passes through the neural network) and space (as those tile-like receptive fields examine each section of a photo). Applying these methods to real problems, as Zitnick’s students do during the workshops, is a process of trial, error and intuition—kind of like manning the mixing console in a recording studio. You’re not at a physical console. You’re at a laptop, sending commands to machines in Facebook data centres across the Internet, where the neural networks do their training. But you spend your time adjusting all sorts of virtual knobs—the size of the dataset, the speed of the training, the relative influence of each node—until you get the right mix.

the flexibility needed to achieve high performance for convolution computation. Myriad also comes with a development kit that includes dedicated software libraries to support sustained performance on matrix multiplication and multidimensional convolution.

Start-up Graphcore proposes to handle deep learning with a so-called intelligent processing unit (IPU)—a graph processor that can manage both training and inference on the same architecture, and eventually across multiple form factors (server and device) too. The chip is expected to get ready for early usage by year-end.

According to the company, “This same architecture can be designed to suit both training and inference. In some cases, you can design a piece of hardware that can be used for training, then segment that up or virtualise it in a way to support many different users for inference or even different machine learning model deployments. There will be cases when everything is embedded, for instance, and you need a slightly different implementation, but it’s the same hardware architecture. That’s our thesis—one architecture, the IPU, for training, inference, and

different implementations of that machine that can be used in servers, cloud, or at the edge of the network.” That would be the ultimate thing to wish for!

AI seems so real

At one time, the term ‘AI’ was associated only with robots, but now it is everywhere—from security cameras to cars and enterprise applications.

In Pittsburgh, USA, AI is helping solve traffic woes. Speaking at a White House Frontiers Conference,

Carnegie Mellon University professor of robotics Stephen Smith said that traffic congestion costs the U.S. economy \$121 billion a year, mostly due to lost productivity, and produces about 25 billion kilograms of carbon dioxide emissions. The AI-based smart traffic management system piloted in the city has reduced travel time by 25 per cent, idling time by over 40 per cent and emissions by 21 per cent. Unlike conventional traffic lights that have pre-programmed



Surtrac is an intelligent approach to traffic management, implemented in Pittsburgh, USA